

Introduction

Surgery offers the best chance of cure for oesophageal cancer, but it is among the most demanding operations in oncology. Complications are common, they shorten survival, and they are **most often pulmonary**. Since 2007, Amsterdam UMC has run a preoperative screening and prehabilitation pathway in which a physiotherapist judges fitness and assigns a risk level. Its value rests on one question: **how accurately can complication risk be read from the data that routine care already collects?**

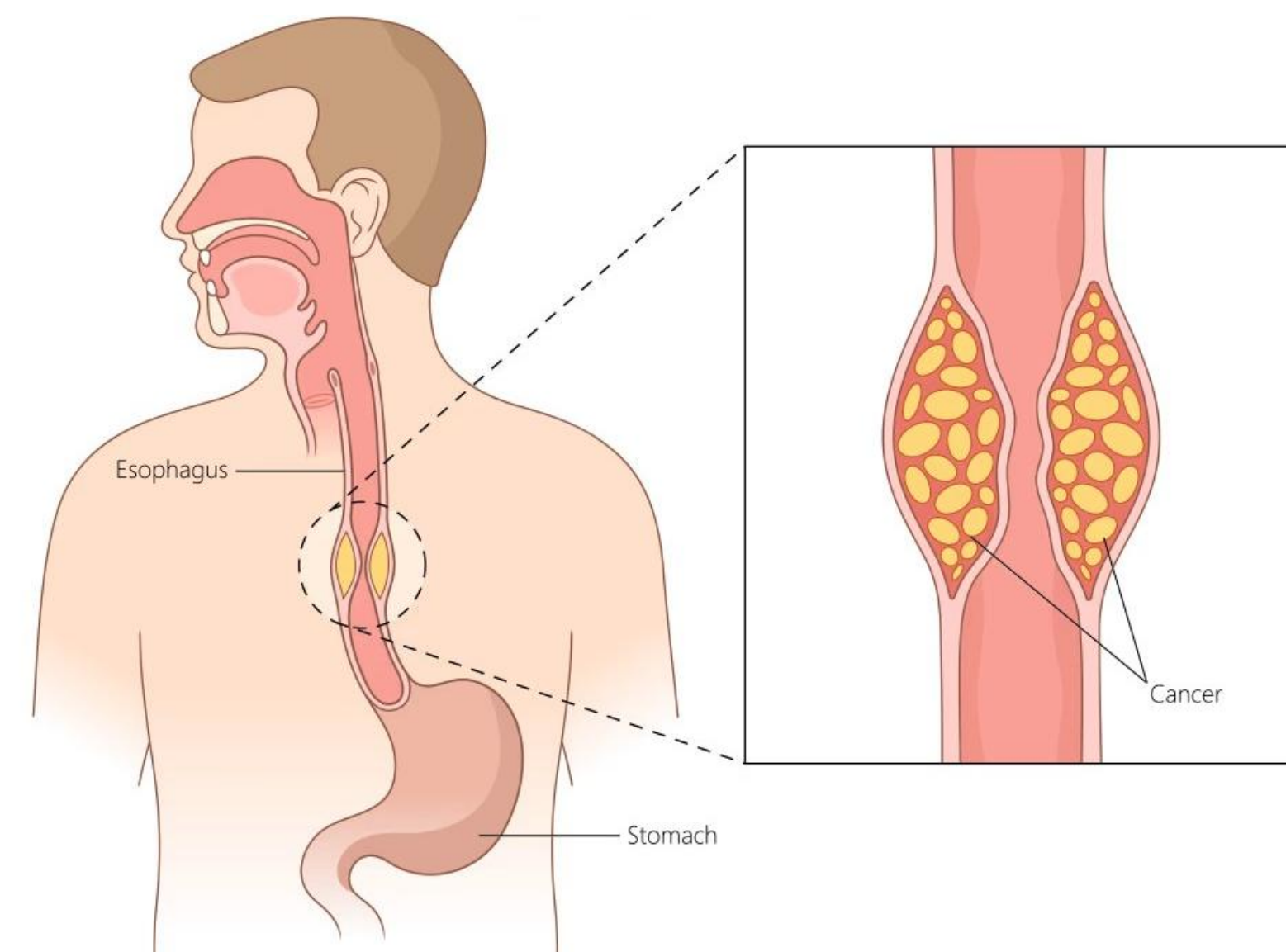


Figure 1. Oesophageal cancer. Tumours arise in the lining of the oesophagus, between throat and stomach; resection is curative but high-risk, with complications most often pulmonary.

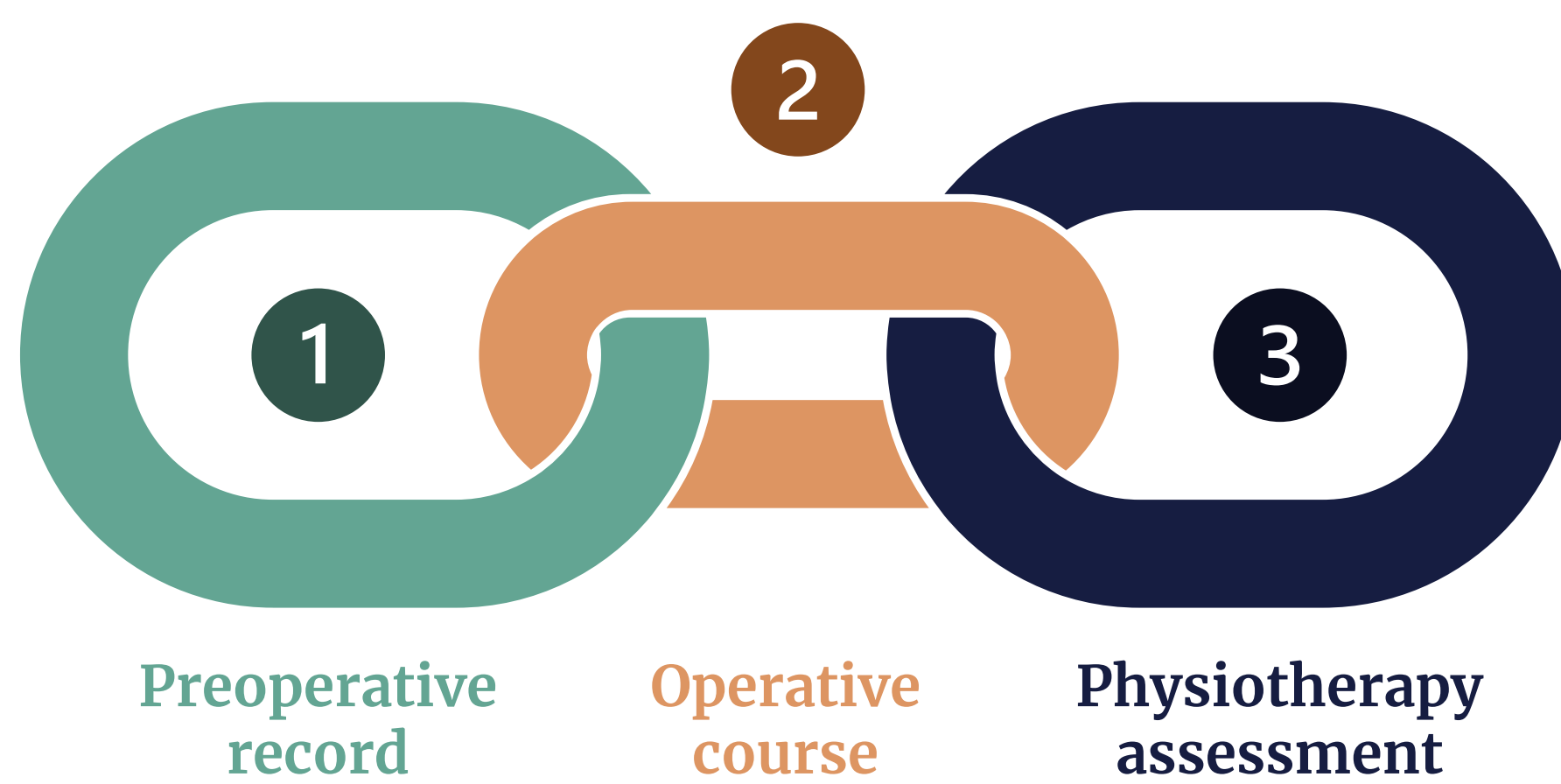
Research Question

How far can complications be predicted from routine information, and does the operative course or physiotherapy assessment raise that ceiling?

Four sub-questions, one per data layer:

- SQ1.** How well do preoperative factors predict the principal outcomes?
- SQ2.** Which factors carry the predictive signal, and are they clinically plausible?
- SQ3.** Does the operative course add value beyond the preoperative record?
- SQ4.** Does the physiotherapy assessment, including ICF¹ functioning from notes, add value beyond surgery?

Approach



- 1. Preoperative record** (before surgery). ASA² class, Charlson index, tumour stage, histology, and neoadjuvant therapy.
- 2. Operative course** (during surgery). Surgical approach, anastomosis type, and conversion to an open procedure.
- 3. Physiotherapy assessment** (at screening). A fitness-based risk conclusion plus ICF functioning, such as exercise tolerance, coded from clinical notes.

Methods

- Outcomes.** Five binary outcomes: major complications (Clavien-Dindo IIIb+)[2], prolonged stay (> 7 days), pulmonary complications, reintervention, and textbook outcome[3].
- Models.** A penalised logistic model, compared against an ASA-only baseline, a lasso, and a random forest.
- Validation.** Stratified 5-fold cross-validation reporting discrimination (AUC), calibration, and decision-curve net benefit, following the TRIPOD+AI guideline[4]. Complete-case, with at least about ten events per predictor.

Data and Cohort

Dutch Upper Gastrointestinal Cancer Audit (DUCA)[1]: 1,022 oesophageal patients, Amsterdam UMC, 2016 to 2025. Median age 67, 79% male, median Charlson 1, mostly ASA 2 to 3, 95% neoadjuvant. 516 linked to a physiotherapy assessment, 440 with a usable conclusion.

Characteristics	No	Yes
ASA 3+	29%	47%
COPD ³	9%	18%
Cervical anastomosis	20%	34%
Neoadjuvant	96%	89%

Table 1. Characteristics by major-complication status. Complicated patients were more often ASA 3+, COPD, and cervical anastomosis.

Results

Figure 2. Cross-validated AUC, preoperative (dark) versus perioperative (yellow). AUC 0.50 is a coin flip and 1.00 is perfect, so 0.60 is weak discrimination. Only major complications gain, 0.594 to 0.638.

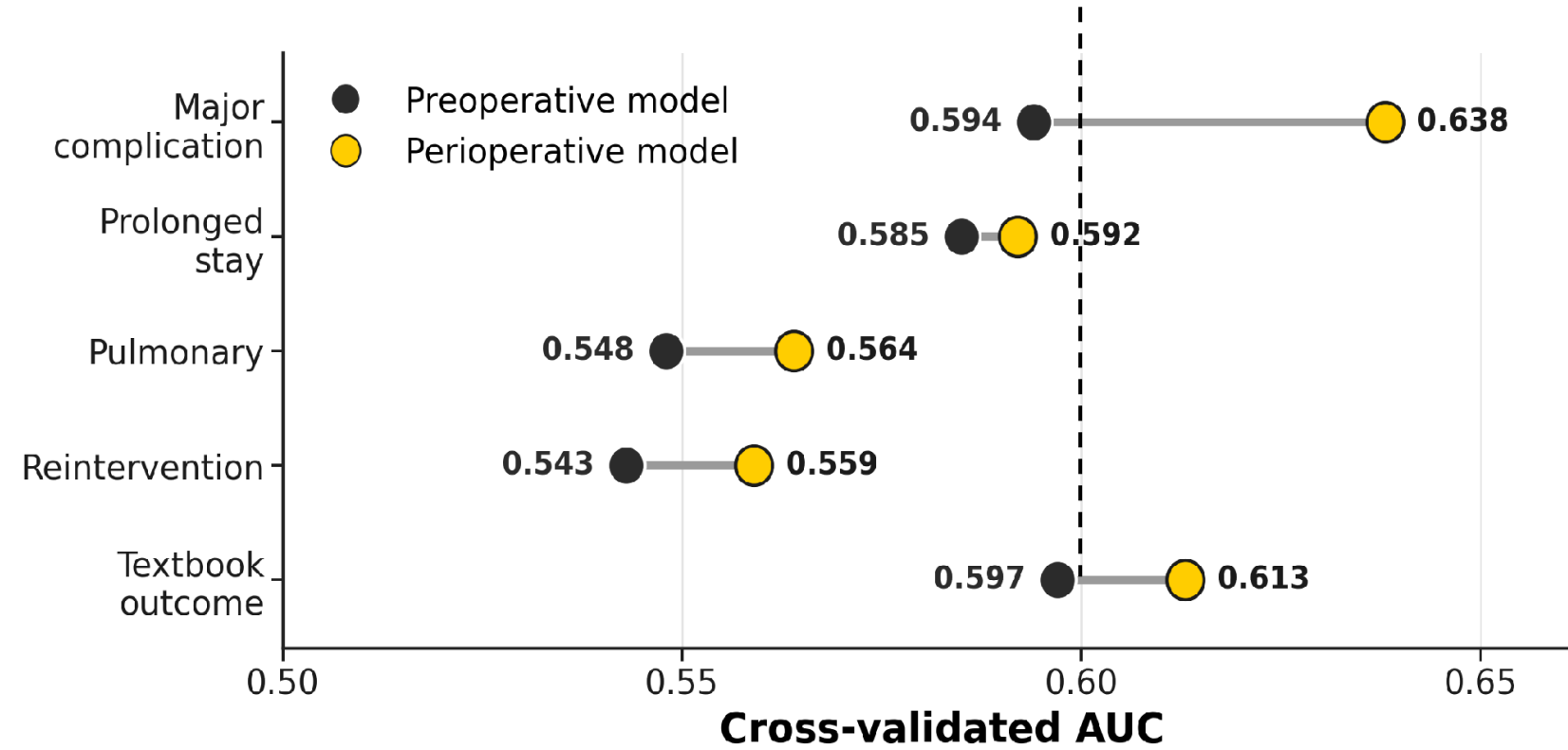


Figure 3. Adjusted odds ratios for major complications. Conversion to open surgery was the strongest predictor (OR 3.44, 95% CI 1.35 to 8.78). As conversion was rare, the confidence interval is relatively wide.

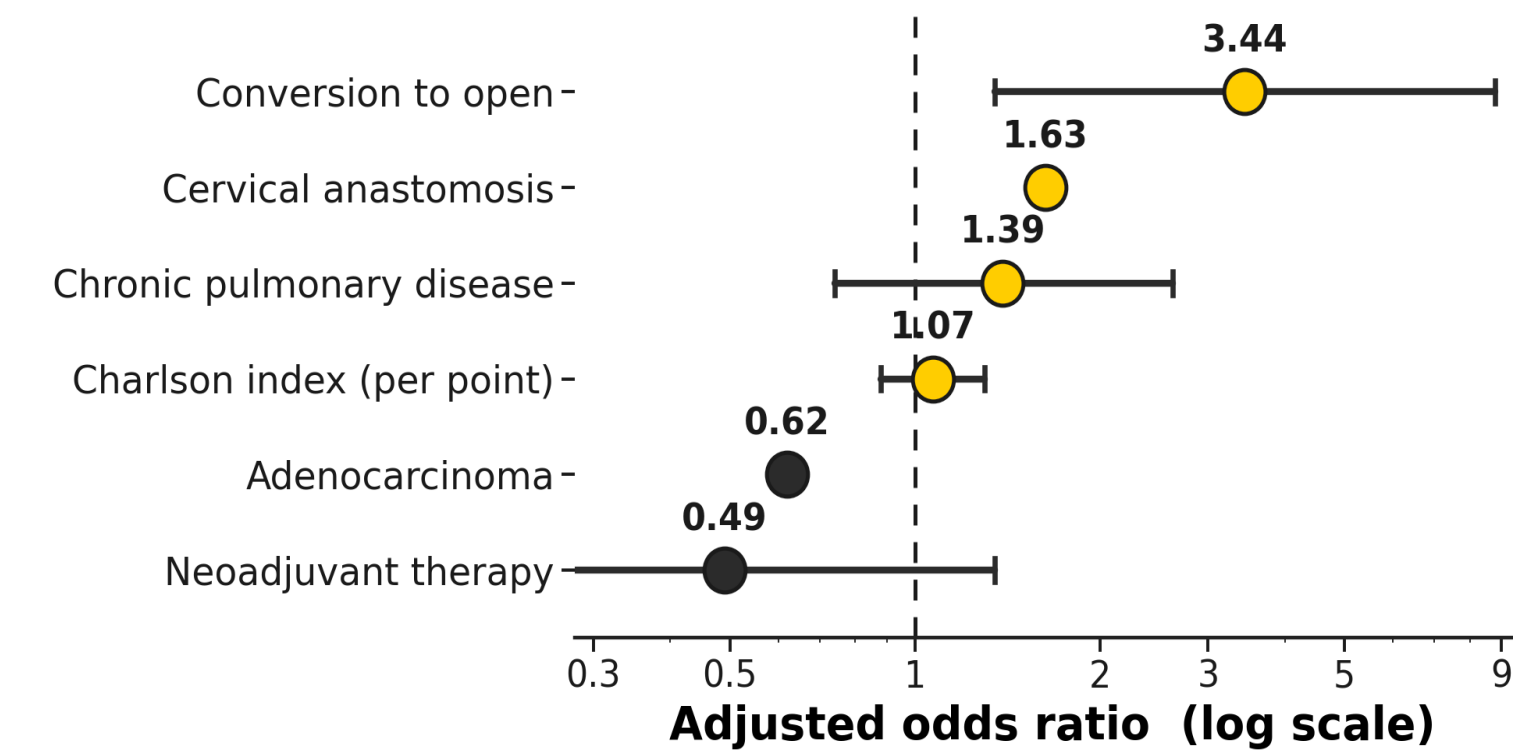
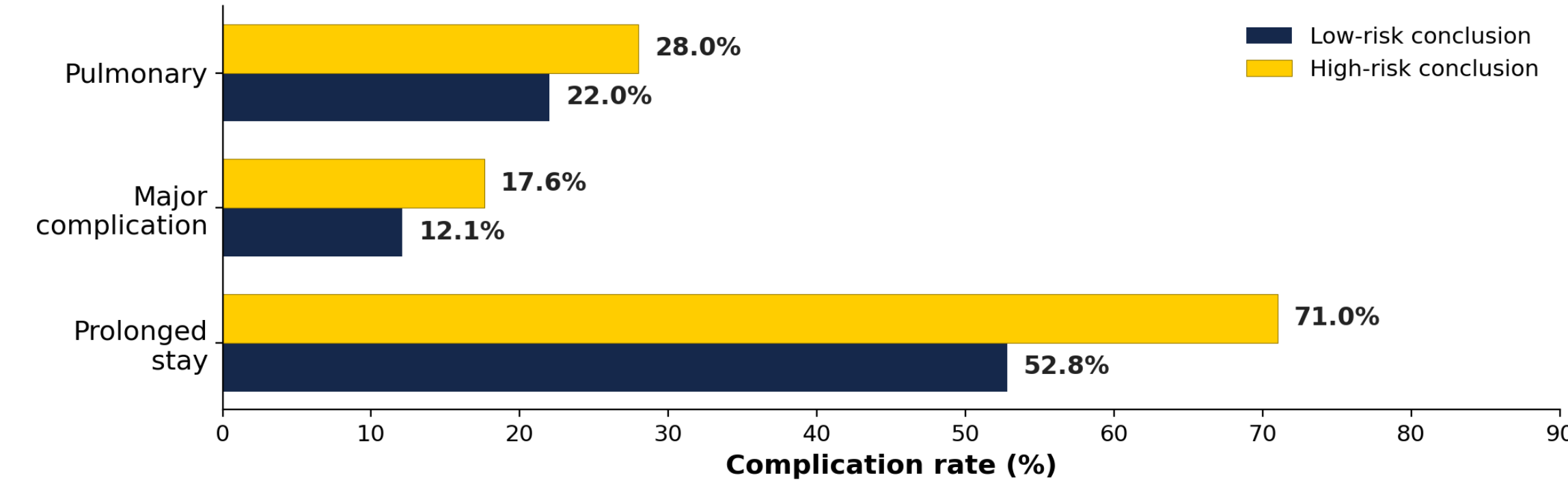


Figure 4. Complication rate by physiotherapist conclusion. High-risk patients fare worse on every outcome, but the assessment adds no discrimination beyond surgery.



Conclusions

Routine data hits a **prediction ceiling near AUC 0.60**; **the surgery itself, not the preoperative workup**, drives short-term complications.

- The operative course helps only for major complications**; conversion to open is the strongest driver, though rare and imprecise.
- The physiotherapy assessment and ICF functioning track risk but add no discrimination beyond surgery.**
- Flexible machine learning does not beat logistic regression.** Calibration: textbook 0.99, major 0.81, prolonged stay 0.82, pulmonary 0.48 (overconfident).
- Implication.** Use these tools to target prehabilitation, not to triage.

Discussion

- 1. Surgery drives the ceiling.** Short-term complications come from the operation itself, not the workup, matching the literature[5].
- 2. A measured limit.** Single-centre and complete-case; the assessment may shape the very risk it predicts.